

Problem 2.1

$$\xrightarrow{-3} \begin{bmatrix} 2 & 3 & | & 5 \\ 6 & 15 & | & 12 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ -3 & 1 & | & 1 \end{bmatrix} = E_{21}$$

$$\longrightarrow \begin{bmatrix} 2 & 3 & | & 5 \\ 0 & 6 & | & -3 \end{bmatrix}$$

$$6y = -3 \longrightarrow y = -\frac{1}{2}$$

$$2x - \frac{3}{2} = 5 \longrightarrow x = \frac{13}{4}$$

Problem 2.2

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad \text{Not}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$